MASSACHUSETTS INSTITUTE OF TECHNOLOGY

2026 BEAVER WORKS SUMMER INSTITUTE

July 6, 2026

Welcome to the 2026 MIT Beaver Works Summer Institute!!

This summer you will be part of a class of more than 400 talented young people from across the United States. We hope that you are ready for a transformational learning experience over the next four weeks! Our program is quite rigorous and will provide you with a good understanding of what it means to be an engineer and scientist, as well as introduce you to career opportunities and current research in the engineering and science professions.

In addition to the technical work you will be performing as part of your classes, you will also attend daily guest lectures and seminars, and have assignments to work on at night. Your teacher assistants (TA's) and instructors will be available to assist you in the classroom.

Please listen carefully during the First Day Orientation. Do not be shy about raising your hand and asking questions. Please read over all program materials provided, paying particular attention to your class schedule and syllabus. You will be responsible for being where you need to be, on-time, and ready to participate. Listen to your instructors and program staff. Be respectful of them and your fellow students. Everyone comes from different backgrounds and has different levels of experience, so please practice tolerance and patience with your fellow students.

Thank you!

Sincerely,

The 2026 MIT BWSI Staff

10 YEARS BEAVER WORKS
MIT Lincoln Laboratory MIT School of Engineering

Code of Conduct for Beaver Works Summer Institute

Each participant in the MIT BWSI Program has a responsibility to add positively to the high standards of the academic, social, and scholarly environment. Participants in the program shall strive to ensure that all actions are consistent with their fundamental responsibility to contribute freely to the intellectual core of the program. Students who violate the rules and/or code of conduct are subject to punishment up to and including immediate dismissal from the program. Your acceptance into the program shows our confidence that you will conduct yourself with the highest regard for moral, legal, ethical, and academic standards. Dismissal from this program is at the sole discretion of the administration of the program and does include dismissal for anything that is disruptive to the program. This program is one of excellence. In continuing that MIT tradition, we require the following.

1. Students shall maintain clarity of mind, body, and spirit. Consuming, inhaling, transporting, or selling of alcoholic beverages or illegal drugs by students is prohibited and will lead to immediate dismissal from the program. Students are not permitted to be in any room or setting where alcohol is served. Students shall not engage in inappropriate activities with other program participants.

2. Students are required to maintain peaceful, encouraging, and socially and academically constructive relationships with other participants. Any activities or interactions that compromise any student's personal or social well-being are grounds for dismissal from the program.

3. Students will add constructively to the academic environment and their surroundings. Defacement or destruction of the classroom, dormitory, or other MIT property will seriously jeopardize the future of this program and are grounds for immediate dismissal. Students whose behavior is judged by administration to be disruptive to the program may be sent home.

4. Students will actively engage in their environment and diligently complete their assigned tasks. All activities both academic and non-academic, including class attendance and completion of homework assignments, are required in this program (unless otherwise indicated on the schedule). The administrative staff may, at its discretion, elect to send a student home if his or her academic responsibilities are judged to have been neglected.

5. Students will show a commitment and focus to uninterrupted participation in the program.

6. Students will behave appropriately in online activities, you are a representation of MIT. No inappropriate backgrounds or names can be used in any online activities that are BWSI related.

RULES FOR ONLINE CONDUCT FOR BWSI

The listing below provides only summaries of the rules for Online conduct for BWSI students; staff, parents and/or visitors.

Don't violate the privacy of other users.

Don't misuse the intellectual property of others.

Don't use BWSI platform(s) to harass anyone in any way.

Don't misuse electronic communications and collaboration services (including, without limitation, age appropriate virtual backgrounds).

This program is for students to learn, parents are not allowed into any virtual portions of the program unless specifically invited by the BWSI staff.

Rules of Use

All BWSI students; staff and/or visitors are expected to follow these rules. ***Violations of the rules can subject the offender to Institute disciplinary proceedings, dismissal from the program, and, in some cases, civil or criminal prosecution.***

Note: Laws that apply in "the real world" also apply in the "virtual" computer world (including MITnet). Laws about libel, harassment, privacy, copyright, stealing, threats, etc. are *not* suspended for computer users, but apply to all members of society whatever medium they happen to be using: face-to-face, phone, *or* computer. Furthermore, law-enforcement officials are more computer-savvy than ever, and violations of the law in "Cyberspace" are vigorously prosecuted.

Similarly, Institute policies (as described in MIT's Policies and Procedures, for example) also apply to online postings.

Artificial Intelligence (AI) Policy

AI is a tool, and students are encouraged to explore its capabilities and limitations. However, the purpose of this course is to develop your understanding of the subject matter and the ethical use of technology—not to rely on AI to generate answers.

Submitting AI-generated text, code, images, videos, or other content as your own work is considered academic misconduct. If AI tools are used for an assignment, you must clearly disclose the tool used, explain how it was used, and include the prompts that generated the content.

Students are responsible for ensuring that all submitted work accurately reflects their own learning and understanding. Additional assignment-specific AI guidelines may apply.

Assuring Ethical Use of the System

Along with the many opportunities that BWSI provides the community to share information comes the responsibility to use the system in accordance with MIT standards of honesty and personal conduct. Those standards, outlined in Section 13.2 of MIT's Policies and Procedures, call for all members of the community to act in a responsible, professional way.

Don't violate the privacy of other users.

The Electronic Communications Privacy Act (18 USC 2510 *et seq.*, as amended) and other federal laws protect the privacy of users of wire and electronic communications.

As Section 11.1 of MIT's *Policies and Procedures* notes, "Invasions of privacy can take many forms, often inadvertent or well-intended." All staff, students and/or visitors should make sure that their actions don't violate the privacy of other users, if even unintentionally.

Some specific areas to watch for include the following:

Don't make any personal information about individuals publicly available without their permission. This includes both text and number data about the person (biographical information, phone numbers, etc.), as well as representations of the person (graphical images, video segments, sound bites, etc.) For instance, it is *not* appropriate to include a picture of someone on a World Wide Web page without that person's permission. (Depending on the source of the information or image, there may also be copyright issues involved; cf. Rule 4).

Don't create any shared programs that secretly collect information about their users. (This means, for example, that you may not collect information about individual users without their consent.)

Don't misuse the intellectual property of others.

MIT faculty, students, and staff produce and consume a vast amount of intellectual property, much of it in digital form, as part of our education and research missions. This includes materials covered by the patent, copyright, and trademark laws, as well as license or other contractual terms.

Members of the MIT community also avail themselves of a wide variety of entertainment content that is available on the Internet, most of which is protected by copyright or subject to other legal restrictions on use.

All users need to ensure that their use of all these protected digital materials respects the rights of the owners.

Digital materials that may be covered by this rule, without limitation, are:

- Data
- E-books
- Games
- Journals and periodicals
- Logos

- Movies
- Music
- Photographs and other graphics
- Software
- Textbooks
- Television programs
- Other forms of video content

You should assume that all materials are subject to these legal protections, and may have some restrictions on use. Ease of access, downloading, sharing, etc., should not be interpreted as a license for use and redistribution.

Don't harass anyone in any way.

"Harassment," according to MIT's Policies and Procedures (Section 9.5), is defined as:

"...any conduct, verbal or physical, on or off campus, which has the intent or effect of unreasonably interfering with an individual or group's educational or work performance at MIT or that creates an intimidating, hostile or offensive educational, work or living environment... Harassment on the basis of race, color, gender, disability, religion, national origin, sexual orientation or age includes harassment of an individual in terms of a stereotyped group characteristic, or because of that person's identification with a particular group."

The Institute's harassment policy extends to the networked world. For example, sending email or other electronic messages which unreasonably interfere with anyone's education or work at MIT may constitute harassment and is in violation of the rules of BWSI.

Any member of the MIT community who feels harassed is encouraged to seek assistance and resolution of the complaint. Contact BWSI staff for help with the situation.

Don't misuse electronic communications and collaboration services.

Some members of the BWSI community access similar, or additional, 3rd party services on the Internet.

Users of all such services have a responsibility to use these services properly and to respect the rights of others in their use of these services, and in accordance with published terms of service.

Users may not use these services in violation of any applicable law.

All relevant MIT policies apply to the use of these services, but in particular:

- Any use that might contribute to the creation of a hostile academic or work environment is prohibited, (including, without limitation, age appropriate virtual backgrounds)
- Any commercial use not required for coursework, research or the conduct of MIT business is prohibited,

- Any non-incidental personal use such as advertisements, solicitations or promotions is prohibited [Note: some services exist on campus that have been designed for buying, selling and exchanging items within the MIT community, and those are allowed].

Any content posted to a service that is inconsistent with these rules, as well as unsolicited mail from outside of MIT (e.g., spam), may be subject to automated interception, quarantine and disposal.

This policy was last updated in June 2026.

Beaver Works Summer Institute (BWSI)

Student Program Details

This document provides information for participants of the MIT Beaver Works Summer Institute (BWSI) Program. Please use this document as a reference.

Any questions should be directed to program staff in person or via email at bwsi-admin@mit.edu.

In the case of an urgent matter, call program staff at 617-744-9398. This number will not be monitored until July 6, 2026.

Virtual Program Expectations:

All students are required to be in the virtual program with video and sound daily by 5 minutes before their class starts. If a student does not show up with their video enabled and does not enable it upon request from instructors they will be asked to leave the program. Parents will be notified if a student is not following the rules of the program. Students must relay to any instructors immediately if there are issues.

In-Person Program Expectations:

All students are required to arrive at the program 5 minutes before their class starts. Students need to come prepared with their laptop ready to use. If you do not have a working laptop let the staff know prior to the start of the program. If a student is habitually late will be asked to leave the program. Parents will be notified if a student is not following the rules of the program. Students must relay to any instructors immediately if there are issues.

Tardy/Absence Policy

If you will be late, you must contact your course instructors via slack, discord or whatever course communication has been set up for your course as soon as you course instructors to avoid possible termination from the program. When emailing please title the message with your full name, course name and cc your parent(s) or guardian(s). Unexcused tardiness or absence from class will result in immediate termination, we will notify parents immediately if a child is asked to leave the program. If a student is sick, or for some reason feels like they cannot make it to class or a scheduled event, they will be required to contact the MIT BWSI Program Manager at bwsi-admin@mit.edu and you course instructors via slack, discord or whatever course communication has been setup for your course as soon as possible to avoid possible termination from the program.

Schedule

Students may find this program to be more challenging than high school. Students spend about many hours per day in classes and other mandatory activities. Students may expect to spend an average of two to three hours after class working on their projects.

The first day is Monday July 6th, 2026. This is orientation and a welcome day. All students are required to attend July 6th! This is mandatory.

Virtual courses are scheduled Monday through Friday, 10AM – 6PM EST. All students are expected to be in their virtual classroom by 9:55 AM EST each day ready to work.

In-Person courses are scheduled Monday through Friday 9AM – 5PM EST. All students are expected to be in their classrooms on MIT campus by 8:55 AM each day ready to work. Unless otherwise indicated on the schedule, this program requires attendance at all program activities, both academic and non-academic.

Lunch

Virtual lunchtimes will be dependent on your course schedules and will be explained to you by your instructors. During lunch time you can break from the virtual session but are required to be back in the session 5 minutes before the end of the break.

Attire

All classes and other program events are casual attire and we are asking that students be respectful of other students in this virtual and in-person environment and dress appropriately. While casual, remember that you are in a professional environment. Revealing clothing such as short shorts, exposed midriffs, or spaghetti straps are not acceptable.

Laptop Computer and Class Supplies

Students must provide their own laptop computer for use in the program, and for homework. Some courses will be sending hardware kits to students as part of the curriculum. Due to the accelerated nature of BWSI, it is very important that care is taken with the materials sent, as there may not be time to replace broken or lost parts before the end of the course. If you have requested equipment from us we will be in touch soon for the loan of this equipment.

As a participant in BWSI, you will be provided an MIT email (aka Kerberos ID), this is so that you can authenticate for mit.zoom.us meetings. Please do not hesitate to ask program staff if you have any questions. Your safety is paramount, so please ask first!

Thank you,

The BWSI 2026 Program Staff

Beaver Works Summer Institute
First Day Program Agenda - SGAI

Monday, July 6, 2026

Remote Attendee - Join Zoom Meeting:

zoom link Password: 004178

9:00 am to 9:30 am	Getting settled, playing Kahoot
9:30 am to 9:45 am	Dr. Robert Shin, Principal Staff, MIT Lincoln Laboratory, Welcome to BWSI
9:45 am to 10:00 am	Scott Van Broekhoven, Asst. Division Head, MIT Lincoln Laboratory, Welcome to BWSI
10:00 am to 10:15 am	Professor Sertac Karaman, Director of the Laboratory for Information and Decision Systems (LIDS), MIT, Welcome to BWSI
10:15 am to 10:45 pm	Joel Grimm, Lisa Kelley, and Sigrid Flender overview of BWSI Program and expectations
10:45 am to 11:00 am	Quick Break
11:00 pm to 12:15 pm	Presentation: Resource and Techniques for Managing Mental Health with Charmain Jackman
12:15 pm to 1:30 pm	Lunch and Socializing
1:30 pm to 5:00 pm	Released to Class

10 YEARS

BEAVER WORKS
MIT Lincoln Laboratory MIT School of Engineering

Beaver Works Summer Institute

Daily Schedule - SGAI

Generic Daily Virtual Schedule

9:55 am	Students Arrive in their Classroom Meeting: https://mitll.zoomgov.com/j/1658996778?pwd=VhA5TkxJ6UDOvLC1VpKOvfhlgEmLtJ.1
10:00 am	Class Start – Lectures and Coursework
12:30 pm	Daily Technical Seminar – see dashboard for schedule (this is the same link every day) Meeting Link and Information (EST): https://mit.zoom.us/j/94570118198?pwd=on6DaGgS6mPNINMjYVUDYfPjNmhl2M.1 Password: 004178
1:30 pm	Classes Resume – Lectures and Coursework
6:00 pm	Class Ends

SGAI Course Syllabus 2026

This schedule will be frequently revised during the course. The version below will give you a good sense of the topics and flow, but expect individual lecture dates to move around a lot. Show up every day and you'll be fine.

Attendance Pragmatics

- **Class** is every weekday **10am-6pm East Coast Time** (7-3 West Coast). Times in the schedule (below) are in east coast time. Subtract 3 for west coast time.
- There may be additional **project team meetings in the evenings and weekends** as needed, but those are flexible and will be determined by your team. We will also provide class time to do development work and complete reading assignments.
- All course lectures, discussions, and teamwork will be remote. Log into the classroom every day for attendance, and you will get further instruction from there. If you have trouble or get lost, ask for help via email or Discord.
- If you need to be **absent**, notify the following people (in advance, if possible; afterwards, if necessary).
  - Lead Instructor: robert.seater@ll.mit.edu
  - BWSI Admins: bwsi-admin@mit.edu
  - Your project team (via Discord)
- If you are **late** by accident, just come as soon as you can (but still come!). These things happen on occasion; just let us know you made it, and it won't be a big deal. We need to keep track of you, but we completely understand that mistakes happen.
- TA's and instructors are here to help; if you need help (big or small), just ask. Any TA can help any project team, but your team will be assigned 1 or 2 TAs that you'll get to know very well.

Instructor Email: bwsi_instructors_sgai@mit.edu

Daily Schedule

- 9:45-9:55 Attendance
- 10-12 Morning block
- 12-1:30 BWSI-wide lunchtime talk (some days) or a break (others); sometimes it runs over
- 1:30-2:00 Attendance & math puzzlers (if the lunch talk runs over, just come back to SGAI class when that's done)
- 2-4 Early afternoon block
- 4-6 Late afternoon block

When in doubt, join the class Zoom or check the class Discord.

Week 1 (introduction & lectures & project ideas)

Day 1 (Monday, July 6th)

- **MORNING**: BWSI-wide orientation (not SGAI-specific; NOT the normal classroom Zoom)
- **AFTERNOON**: tech test, course introduction, serious games overview, Minecraft activity conclusion

Day 2 (Tuesday)

- **MORNING**: lecture spillover, online course discussions□ new number 2a
- **AFTERNOON**: intro to baseline zombie game, CNN intro, Git Intro, teams meet

Day 3 (Wednesday)

- **MORNING**: agile overview, TA advice, reading assignment
- **AFTERNOON**: types of AI, AI classifiers, reading assignment

Day 4 (Thursday)

- **MORNING**: reading groups
- **AFTERNOON**: methods of categorization, game definitions, AI in games, project ideas, team project brainstorming

Day 5 (Friday)

- **MORNING**: serious game methods deep dive
- **AFTERNOON**: more serious game methods deep dive, data science horror story, establish team roles and rhythm, team project brainstorming and assessing risk, optional evening social activity

Weekend

- team brainstorming

Week 2 (lectures & project planning & project end-to-end baselining)

Day 6 (Monday, July 13th)

- **MORNING**: LLMs
- **AFTERNOON**: AI deep dive, code base review, project team dev time

Day 7 (Tuesday)

- **MORNING**: RL overview
- **AFTERNOON**: more RL overview, project team dev time

Day 8 (Wednesday)

- **MORNING**: math of NNs, TA advice (part 2)
- **AFTERNOON**: visualizing information, assigned reading, project team dev time

Day 9 (Thursday)

- **MORNING**: reading groups
- **AFTERNOON**: requirements engineering, project team dev time

Day 10 (Friday)

- **MORNING**: moral frameworks, project team SWOT
- **AFTERNOON**: ethics guest lecture, project team dev time, optional evening social activity

Weekend

- team project dev time

Week 3 (mostly software development)

Day 11 AM (Monday, July 20st)

- **MORNING**: economics, project team dev time
- **AFTERNOON**: code review prep, project team dev time

Day 12 AM (Tuesday)

- **MORNING**: TA presentation, project team dev time
- **AFTERNOON**: project team dev time, assigned reading

Day 13 AM (Wednesday)

- **MORNING**: reading groups, project team dev time
- **AFTERNOON**: into to data analytics, data logging and analytics, project team dev time

Day 14 AM (Thursday)

- **MORNING**: advanced python modules, code reviews
- **AFTERNOON**: code reviews, project team dev time

Day 15 AM (Friday)

- **MORNING**: plan for code freeze, presentation advice
- **AFTERNOON**: project team dev time, optional evening social activity

Weekend

- team project dev time

Week 4 (wrap up project; collect data; presentation prep)

Day 16 (Monday, July 27th)

- **MORNING**: prep for data collection, project team dev time
- **AFTERNOON**: data collection, code freeze

Day 17 (Tuesday)

- **MORNING**: TA presentation, project presentation prep
- **AFTERNOON**: data collection, project presentation prep

Day 18 (Wednesday)

- **MORNING**: project presentation prep, rehearse presentations
- **AFTERNOON**: rehearse presentations

Day 19 (Thursday)

- **MORNING**: TA presentation, record presentations
- **AFTERNOON**: record presentations

Day 20 (Friday)

- **MORNING**: TA presentation, TA presentation
- **AFTERNOON**: fun tangents, post-course pragmatics

Weekend

- BWSI closing ceremony
- SGAI project presentations